

Inferential Statistics



STATISTICS

Objective: To extract actionable insights and Statistics from the different datasets like Zingaro Stone, Gunny bags, Dental Implant, and Physiotherapist to utilize statistical analysis.

Data Description

Foot ball

A physiotherapist with a male football team is interested in studying the relationship between foot injuries and those who are not injured.

Gunny bags

The breaking strength of the gunny bags is described by a normal distribution Mean 5 kg per sq. centimeter and standard deviation of 1.5 kg per sq. centimeter. Key parameters that define the distribution of the breaking strength data.

Zingaro Stone

- The dataset contains 75 total observations (rows).
- There are two primary variables for comparison: Unpolished and Treated and Polished.
- Both variables are stored as float64.
- There is no missing value.

Observations	Variable	Duplicate
75	2	0

Dental Implant

- The dataset contains 90 total observations (rows).
- Independent Variables (Factors): Dentist, Method, Alloy, and Temp
- Dependent Variable (Response): Response
- All five variables, including the response variable, are stored as int64
- There is no missing value.

Observations	Variable	Duplicate
90	5	0

Content

Problem 1

- 1.1 What is the probability that a randomly chosen player would suffer an injury?
- 1.2 What is the probability that a player is a forward or a winger?
- 1.3 What is the probability that a randomly chosen player plays in a striker position and has a foot injury?
- 1.4 What is the probability that a randomly chosen injured player is a striker?

Problem 2

- 2.1 What proportion of the gunny bags have a breaking strength of less than 3.17 kg per sq cm?
- 2.2 What proportion of the gunny bags have a breaking strength of at least 3.6 kg per sq cm.?
- 2.3 What proportion of the gunny bags have a breaking strength between 5 and 5.5 kg per sq cm.?
- 2.4 What proportion of the gunny bags have a breaking strength NOT between 3 and 7.5 kg per sq cm.?

Problem 3

- 3.1 Zingaro has reason to believe that the unpolished stones may not be suitable for printing. Do you think Zingaro is justified in thinking so?
- 3.2 Is the mean hardness of the polished and unpolished stones the same?

Problem 4

- 4.1 How does the hardness of implants vary depending on dentists?
- 4.2 How does the hardness of implants vary depending on methods?
- 4.3 What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?
- 4.4 How does the hardness of implants vary depending on dentists and methods together?

Problem 1

1.1 What is the probability that a randomly chosen player would suffer an injury?

 The probability for a player being injured is 61.7% (0.617).

Explanation:

From the data, identify the total number of injured players (145) and the total number of players (235).

Apply the probability formula: The probability of a randomly chosen player being injured is the ratio of the total number of injured players to the total number of players.

1.2 What is the probability that a player is a forward or a winger?

 The probability that a player is a **forward or a winger** is approximately **0.523**.

Explanation:

The total number of players is 235. The total number of forwards is found in the Total is 94. The total number of wingers is found in the Total is 29.

Calculate the probability. The probability of a player being a forward or a winger is the sum of their individual probabilities.

1.3 What is the probability that a randomly chosen player plays in a striker position and has a foot injury?

 The probability of a randomly chosen player plays in a striker position and has a foot injury is 0.191.

Explanation:

The total number of players is 235. The number of players who are both strikers and injured is 45.

The probability of a player being a striker and injured is the number of players 45 who are both divided by the total number of players 235. Is the output of 0.191.

1.4 What is the probability that a randomly chosen injured player is a striker?

 The probability that a randomly chosen injured player is a striker 0.310.

Explanation:

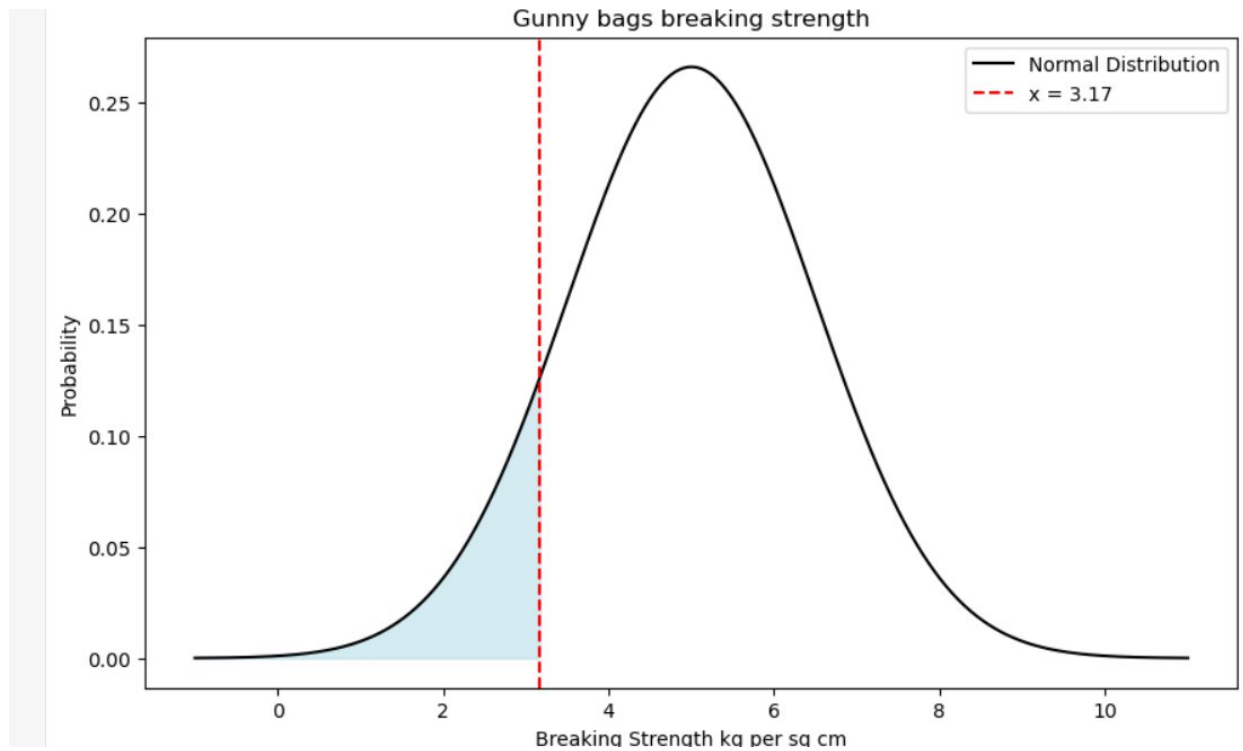
This is a conditional probability problem. The number of injured strikers is 45. The total number of injured players is 145.

The probability that an injured player is a striker is the number of injured strikers 45 divided by the total number of injured players 145.

Problem 2

2.1 What proportion of the gunny bags have a breaking strength of less than 3.17 kg per sq cm?

 The proportion of gunny bags with a breaking strength of less than 3.17 kg/sq. cm is 0.1112 or 11.12%.

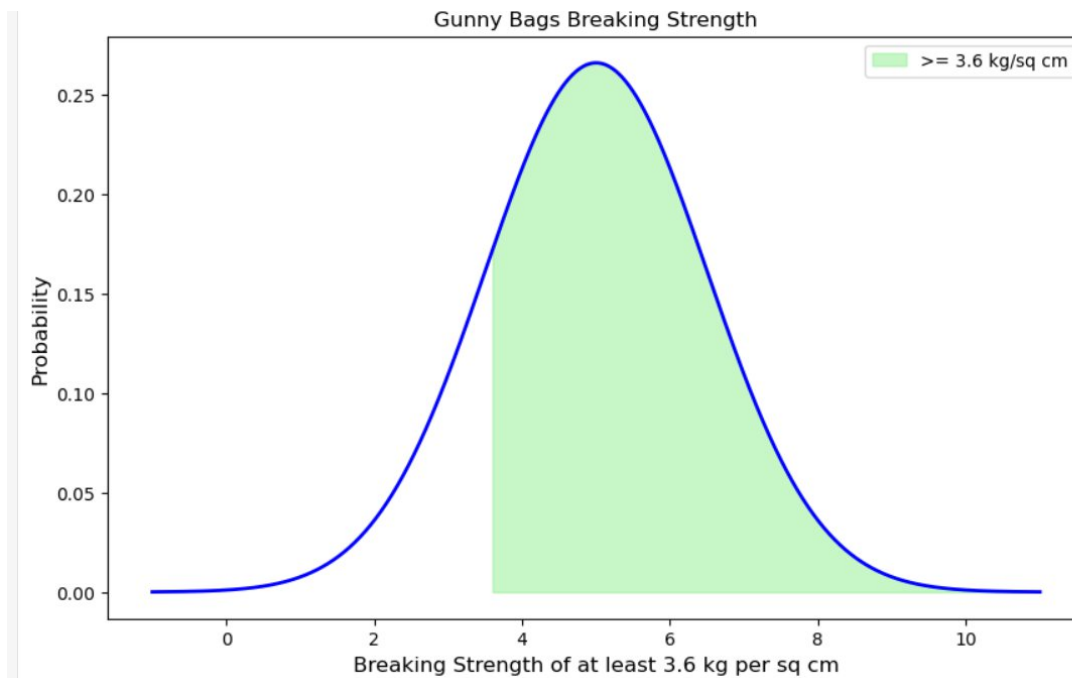


Explanation:

Calculate the z-score and find the proportion where the Mean is 5, Standard Deviation is 1.5 breaking the strength of less than 3.17. The CDF value for $z = -1.22$

2.2 What proportion of the gunny bags have a breaking strength of at least 3.6 kg per sq cm?

The proportion of gunny bags with breaking strength at least 3.6 kg per sq cm is 0.8247 or 82.38%.

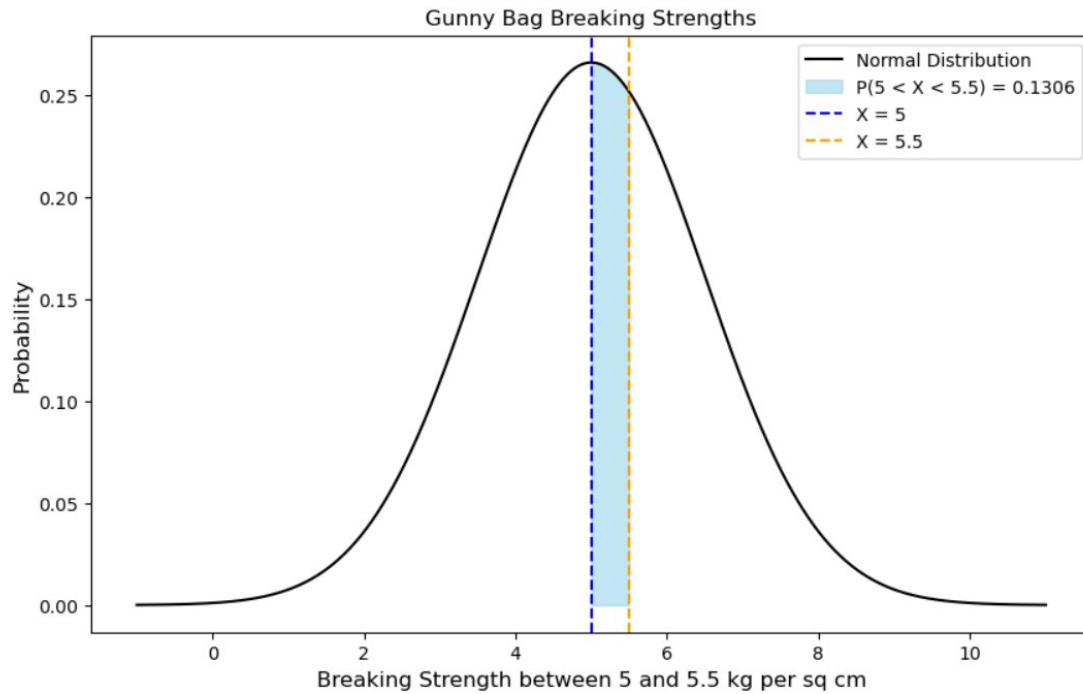


Explanation:

Calculate the Z-score for $X = 3.6$, the cumulative probability for this Z-score is -0.93 or 0.1762 then for the breaking strength of at least 3.6 kg is 82.38%

2.3 What proportion of the gunny bags have a breaking strength between 5 and 5.5 kg per sq cm?

The proportion of gunny bags with breaking strength between 5 and 5.5 kg per sq cm is 0.1306 or 13.06%.

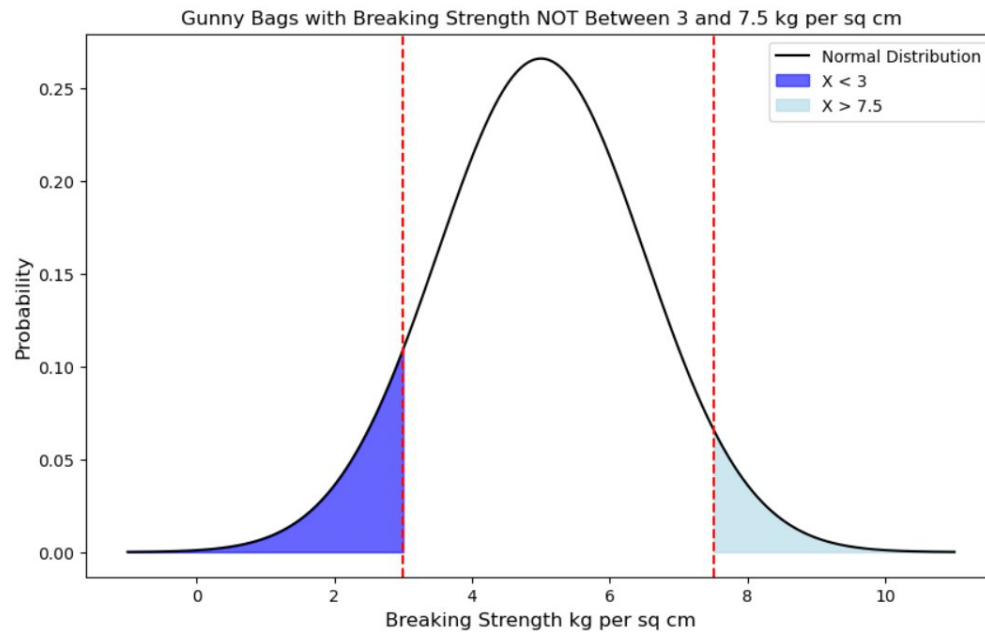


Explanation:

The Z-score for $X_1 = 5$ is 0, and $X_2 = 5.5$ is 0.33. The cumulative probability for $Z_1=0$ is 0.5000, and $Z_2=0.33$ is about 0.6306.

2.4 What proportion of the gunny bags have a breaking strength NOT between 3 and 7.5 kg per sq cm?

The proportion of gunny bags with breaking strength not between 3 and 7.5 kg per sq cm is 0.1393 or 13.93%.



Explanation:

The Z-score for $X_1 = 3$ is -1.33, and $X_2 = 7.5$ is 1.67. The cumulative probabilities of Z_1 are 0.0918 and Z_2 are 0.9525.

3.1 Zingaro has reason to believe that the unpolished stones may not be suitable for printing. Do you think Zingaro is justified in thinking so?

Zingaro is justified in thinking that unpolished stones may not be suitable

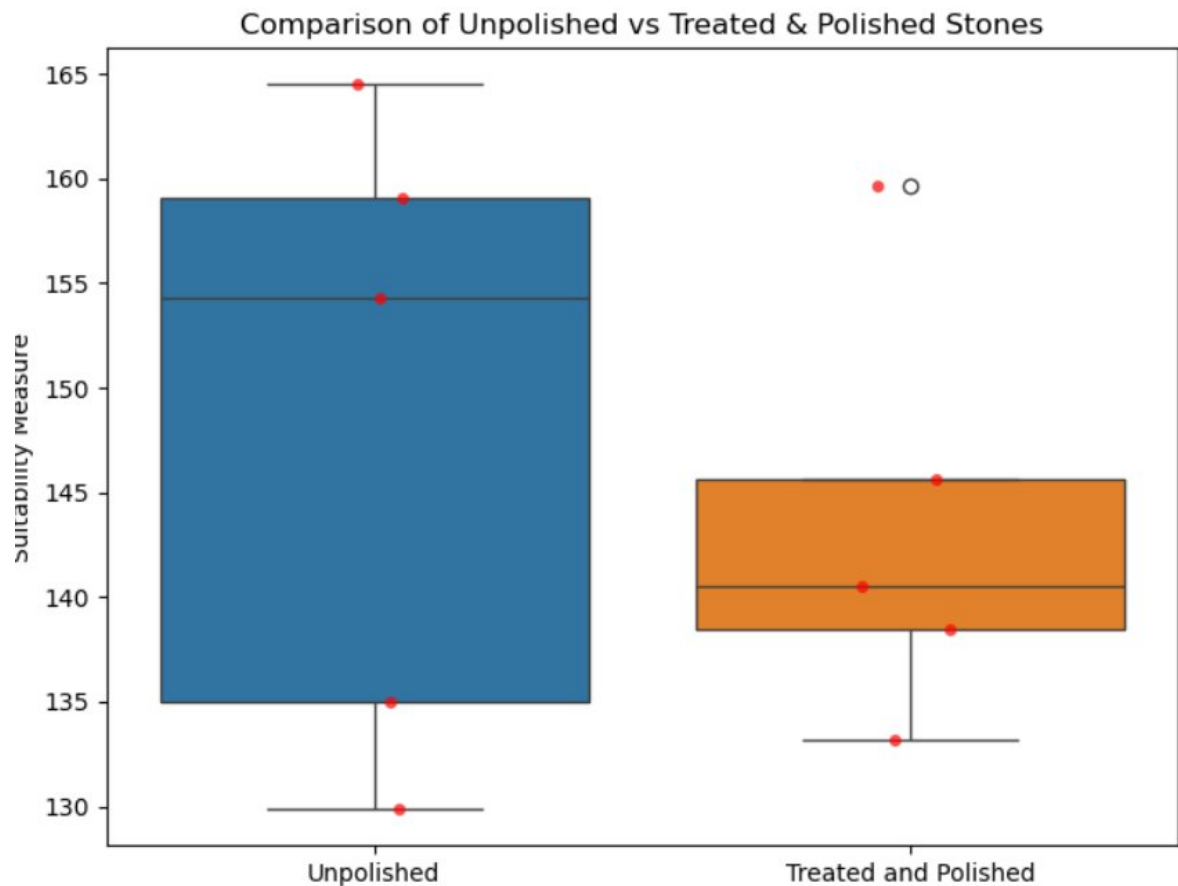
Null Hypothesis(H_0), Alternative Hypothesis (H_1)
level of significance as 5%

T-statistic: 0.4813

P-value: 0.6554

Fail to reject H_1 : believe that the unpolished stones may not be suitable.

Null Hypothesis should be rejected. And Alternative Hypothesis should be accepted.



3.2 Is the mean hardness of the polished and unpolished stones the same?

Mean hardness is different between unpolished and polished stones.

Null Hypothesis (H_0): The mean hardness is the same for both groups.

Alternate Hypothesis (H_1): The mean hardness is NOT the same for both groups.

[Polished Stones] Mean: 142.200

[Unpolished Stones] Mean: 147.167

T-statistic: -0.6563

P-value: 0.5280

Fail to reject H_1 Mean hardness is different between unpolished and polished stones.

4.1 How does the hardness of implants vary depending on dentists?

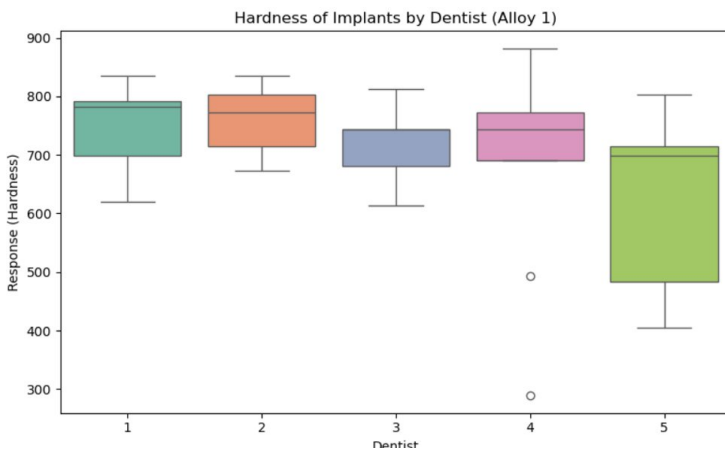
ANALYSIS FOR ALLOY TYPE 1

H0: Mean implant hardness is equal across all Dentists.

H1: At least one Dentist's mean hardness differs.

	sum_sq	df	F	PR(>F)
C(Dentist)	106683.688889	4.0	1.977112	0.116567

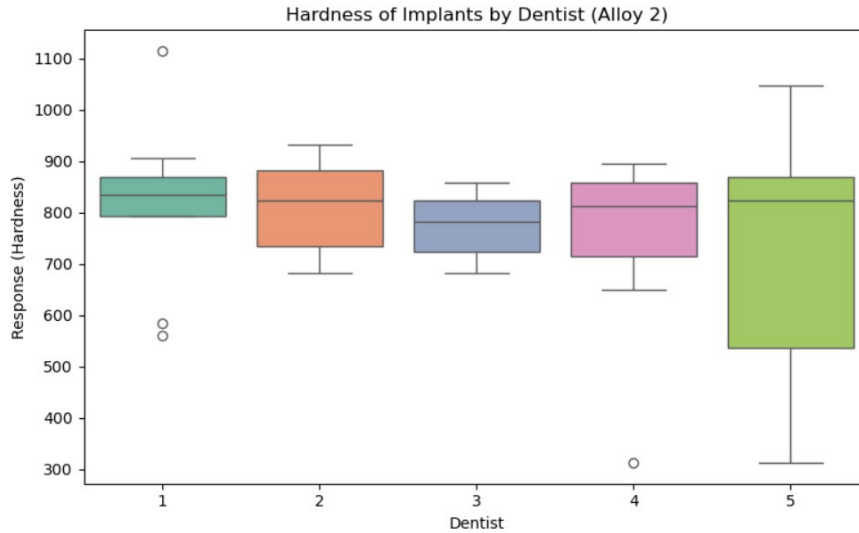
Fail to reject H0 ($p = 0.1166$) → No significant difference in hardness.



ANALYSIS FOR ALLOY TYPE 2

	sum_sq	df	F	PR(>F)
C(Dentist)	5.679791e+04	4.0	0.524835	0.718031

Fail to reject H0 ($p = 0.7180$) → No significant difference in hardness



4.2 How does the hardness of implants vary depending on methods?

Method 1 and Method 3, Method 2 vs Method 3 are differ

ANALYSIS FOR ALLOY TYPE 1

H_0 : The true mean implant hardness is the same across all Methods ($\mu_1 = \mu_2 = \mu_3$).

H_1 : At least one Method's true mean implant hardness differs.

One-Way ANOVA Results:

	sum_sq	df	F	PR(>F)
Method	148472.177778	2.0	6.263327	0.004163
Residual	497805.066667	42.0	NaN	NaN

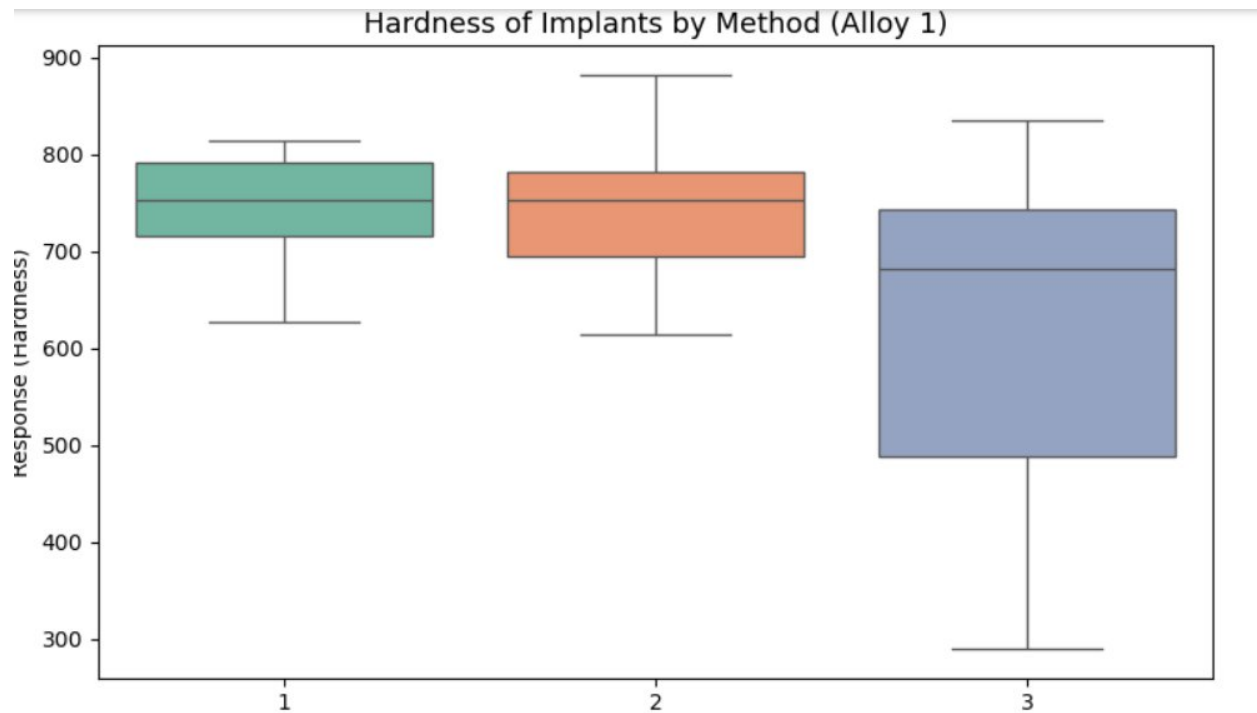
Tukey HSD Results:

Multiple Comparison of Means - Tukey HSD, FWER=0.05

```
=====
group1 group2 meandiff p-adj lower upper reject
-----
1 2 -6.1333 0.987 -102.714 90.4473 False
1 3 -124.8 0.0085 -221.3807 -28.2193 True
2 3 -118.6667 0.0128 -215.2473 -22.086 True
-----
```

Significant differences found between:

- Method 1 vs Method 3
- Method 2 vs Method 3



ANALYSIS FOR ALLOY TYPE 2

H_0 : The true mean implant hardness is the same across all Methods ($\mu_1 = \mu_2 = \mu_3$).

H_1 : At least one Method's true mean implant hardness differs.

one-Way ANOVA Results:

	sum_sq	df	F	PR(>F)
Method	499640.4	2.0	16.4108	0.000005
Residual	639362.4	42.0	NaN	NaN

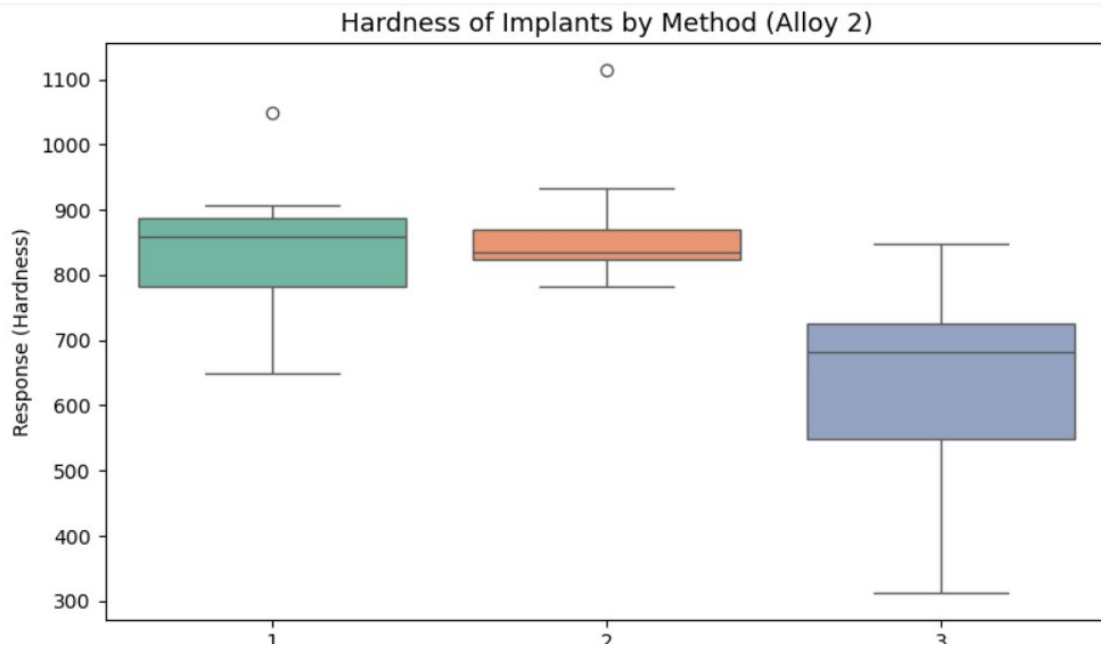
Tukey HSD Results:

Multiple Comparison of Means - Tukey HSD, FWER=0.05

```
=====
group1 group2 meandiff p-adj lower upper reject
-----
1 2 27.0 0.8212 -82.4546 136.4546 False
1 3 -208.8 0.0001 -318.2546 -99.3454 True
2 3 -235.8 0.0 -345.2546 -126.3454 True
-----
```

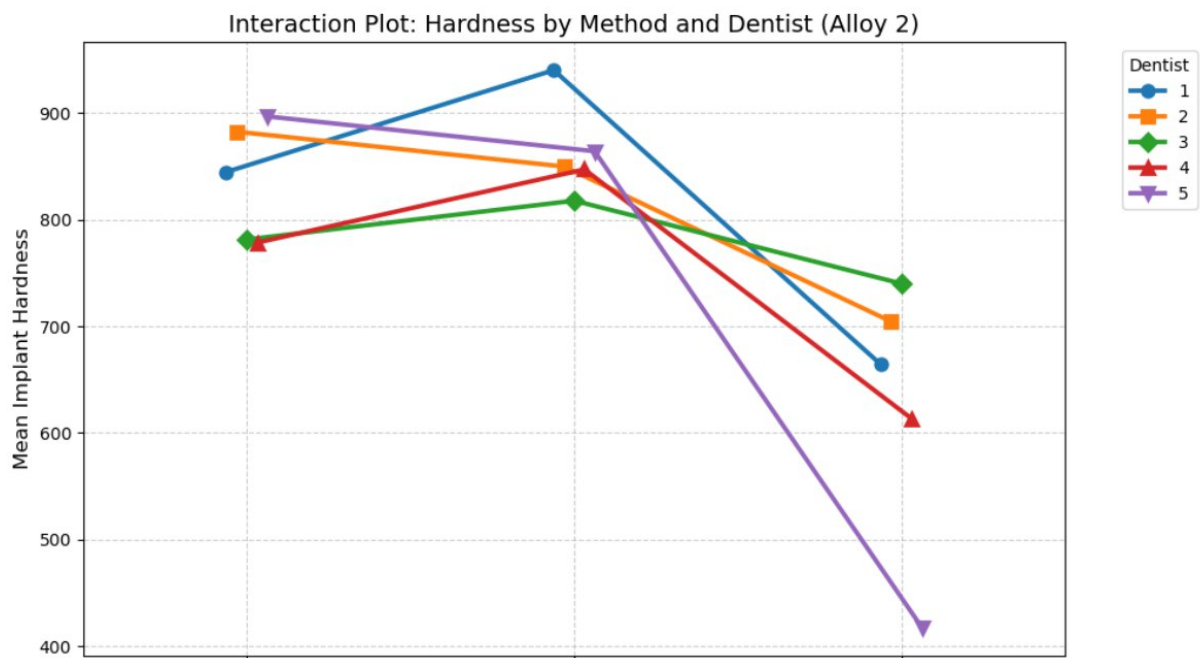
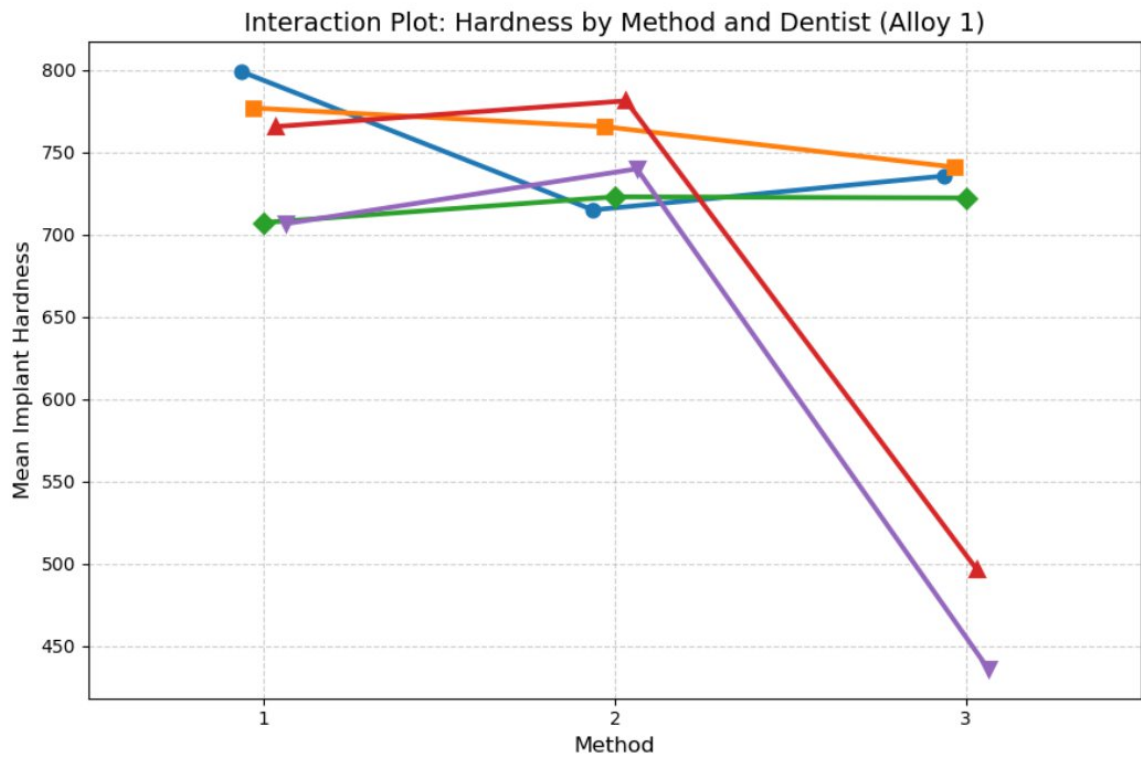
Significant differences found between:

- Method 1 vs Method 3
- Method 2 vs Method 3



Method 1 and Method 3, Method 2 vs Method 3 are differs

4.3 What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?



4.4 How does the hardness of implants vary depending on dentists and methods together?

Alloy 1- The effect of Method on hardness depends on the Dentist.

Alloy 2- The effect of Method is consistent across Dentists.

TWO-WAY ANOVA ANALYSIS FOR ALLOY TYPE 1

- A. Dentist effect: H_0 — Mean hardness same across all Dentists.
- B. Method effect: H_0 — Mean hardness same across all Methods.
- C. Interaction: H_0 — No interaction between Dentist and Method.

Two-Way ANOVA Results (Type II SS):

	sum_sq	df	F	PR(>F)
C(Dentist)	106683.688889	4.0	3.899638	0.011484
C(Method)	148472.177778	2.0	10.854287	0.000284
C(Dentist):C(Method)	185941.377778	8.0	3.398383	0.006793
Residual	205180.000000	30.0	NaN	NaN

Conclusions ($\alpha = 0.05$):

Interaction (Dentist×Method): Significant ($p = 0.0068$)

→ The effect of Method on hardness depends on the Dentist.

Dentist main effect: Significant ($p = 0.0115$)

Method main effect: Significant ($p = 0.0003$)

TWO-WAY ANOVA ANALYSIS FOR ALLOY TYPE 2

Hypotheses:

- A. Dentist effect: H_0 — Mean hardness same across all Dentists.
- B. Method effect: H_0 — Mean hardness same across all Methods.
- C. Interaction: H_0 — No interaction between Dentist and Method

Two-Way ANOVA Results (Type II SS):

	sum_sq	df	F	PR(>F)
C(Dentist)	56797.911111	4.0	1.106152	0.371833
C(Method)	499640.400000	2.0	19.461218	0.000004
C(Dentist):C(Method)	197459.822222	8.0	1.922787	0.093234
Residual	385104.666667	30.0	NaN	NaN

Conclusions ($\alpha = 0.05$):

Interaction (Dentist×Method): Not significant ($p = 0.0932$)

→ The effect of Method is consistent across Dentists.

Dentist main effect: Not significant ($p = 0.3718$)

Method main effect: Significant ($p = 0.0000$)

